Institute for Cognitive Systems
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Neuro-inspired System Engineering - Tutorial 1
Block 2

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Winter Semester 2025

Inertial Measurement Unit (IMU) (41 points)

In this tutorial we work with the data taken from a Sparkfun 9DOF Razor IMU. This IMU integrates a 3D accelerometer, a 3D gyroscope and a 3D magnetometer. The accelerometer measures the linear, centripetal, Coriolis, and the earth gravitational acceleration, the gyroscope measures angular velocities around the coordinate axes, and the magnetometer measures the magnetic flux of the earth magnetic field.
In this tutorial we will learn:

- How to calibrate 3D sensors (accelerometer, gyroscope, magnetometer)

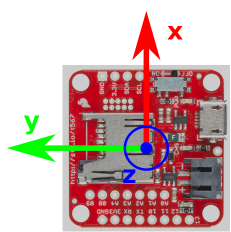

Figure 1: IMU with the axes of the base coordinate system 0. We will use the homogeneous transformations ${}^{\mathrm{acc}}_{0}\mathbf{T}$, ${}^{\mathrm{gyro}}_{0}\mathbf{T}$, and ${}^{\mathrm{mag}}_{0}\mathbf{T}$ to transform the measurements of the sensors to this common coordinate frame.

Preparation – Downloading the tutorial project

All the measurements and Matlab scripts for this tutorial can be found in one common project. You can download this project from TUM Moodle:

```
"https://www.moodle.tum.de/mod/resource/view.php?id=3645451"
```

This project has been extensively tested with `Matlab2020b` and should work out of the box. To source the libraries of the project, run once the script `setup.m`. The sections in the Matlab code that you will modify during this tutorial are marked with a line of question marks `???...???`. To ensure that you are not accidentally modifying code that should not be touched and to avoid breaking your whole project please only modify code between two lines of these question marks. The marked section additionally contains instructions and implementation hints. Please read them carefully. They will help you to save time in your implementations.

1 Accelerometer Calibration (41 points)

The accelerometer calibration incorporates:

- Determining the transformation ${}^{\mathrm{acc}}_{0}\mathbf{T}$ of the accelerometer coordinate frame with respect to (wrt.) the base coordinate frame 0.
- Determining the measurement offset with respect to all three axes.
- Determining the measurement gain with respect to all three axes.

We furthermore assume that the accelerometer's coordinate axes are orthogonal to each other and that the axes are aligned to the sensor module. That is, ${}^{\mathrm{acc}}_{0}\mathbf{T}$ only maps coordinate axes and changes their directions, e.g. all rotations are dividable by 90 deg.

1.1 Determining the transformation between the accelerometer and the base frame (5 points)

To determine the orientations of the accelerometer's coordinate axes with respect to the base frame (coordinate system 0, see Figure 1), we measure the gravitational acceleration by aligning it to the axes of the base frame 0. Then the accelerometer should measure the full magnitude of the gravitational acceleration along one coordinate axis, while the magnitudes along the other axes should be approximately zero.
These measurements are stored in the following Matlab data files:

- negative x-axis along the gravitational acceleration: `accXn.mat`
- positive x-axis along the gravitational acceleration: `accXp.mat`
- negative y-axis along the gravitational acceleration: `accYn.mat`
- positive y-axis along the gravitational acceleration: `accYp.mat`
- negative z-axis along the gravitational acceleration: `accZn.mat`
- positive z-axis along the gravitational acceleration: `accZp.mat`

and can be found in the project folder `data`. For example, in `accXn.mat` we aligned the negative x-axis along the gravitational acceleration, see Figure 2.

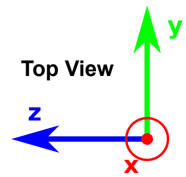

Figure 2: The axes of the base frame for measurement `accXn.mat`.

Open the Matlab script `main_acc_directions.m`. Find the sections that have been marked and require your inputs. This script loads all the measurements of the experiments `accXn.mat` etc. and prints out the matrix ${}^{\mathrm{acc}}_{0}\mathbf{T}$ and the gravitational accelerations for all the experiments.

T.1.1 **(3 points)** Until you modify the script, ${}^{\mathrm{acc}}_{0}\mathbf{T}$ is initialized to identity. Please run the script, inspect the print outs and determine ${}^{\mathrm{acc}}_{0}\mathbf{T}$.

T.1.2 **(2 points)** The script should print out the acceleration measurements wrt. the base frame. Use the marked section to expand the measurements to homogeneous coordinates and transform them to the base frame 0. When ${}^{\mathrm{acc}}_{0}\mathbf{T}$ of **T.1.1** is correct the print outs should match your expectations.

Please submit `main_acc_directions.m` containing your modifications for **T.1.1** and **T.1.2**.

1.2 Calibrating the accelerometer (26 points)

To calibrate the accelerometer we have to find its offset and gain parameters. We will incorporate these gains and offsets into a homogeneous transformation matrix ${}^{\mathrm{raw}}_{\mathrm{acc}}\mathbf{T}$ that transforms the uncalibrated raw measurements of the accelerometer to calibrated measurements in the coordinate frame of the accelerometer.

$$ {}_{\mathrm{acc}}\mathbf{x} = {}^{\mathrm{raw}}_{\mathrm{acc}}\mathbf{T}\ {}_{\mathrm{acc}}\mathbf{x}_{\mathrm{raw}} \tag{1} $$

We determine the gains and offsets of the accelerometer by employing the ellipsoid fitting method introduced in lecture **Block 2 - L1**. The ellipsoid fitting delivers the model matrix $\mathbf{M}$ containing the gains g_i and the rotations of the coordinate axes $\mathbf{R}$

$$ \mathbf{M} = \mathrm{diag}(g_1,\, g_2,\, g_3)\,\mathbf{R} \tag{2} $$

and the offset vector $\mathbf{w}$. Because of the Singular Value Decomposition (SVD) the rotation matrix $\mathbf{R}$ is usually not the identity matrix. Therefore, this rotation has to be compensated for to get ${}^{\mathrm{raw}}_{\mathrm{acc}}\mathbf{T}$.
Open the Matlab script `main_acc_calib.m`. Find the sections that have been marked and require your inputs. This script loads several experiments to provide measurements of the gravitational acceleration in many different poses. The script displays the loaded data, performs the calibration by calling the function `accCalib.m` and finally applies the calibration to the measurements for validation. You will implement the calibration algorithm in the marked section of `accCalib.m`.

T.1.3 **(2 points)** We assume that $\mathbf{R}$ is constrained to rotations that are dividable by 90 degrees, i.e. the axes of the ellipsoid are aligned to the coordinate axes of the accelerometer but could be flipped or mapped differently. E.g. the x-axis of the ellipsoid could be mapped to the negative z-axis of the accelerometer. This constraint simplifies $\mathbf{A}_{\mathrm{fit}}$. Then, how many parameters do you need for $\mathbf{A}_{\mathrm{fit}}$? Please explain.

T.1.4 **(2 points)** What parameters (A, B, $\dots$, K) of the approximated ellipsoid matrix $\tilde{\mathbf{A}}$ become zero? Please explain.

T.1.5 **(8 points)** Implement the calibration algorithm discussed in the previous lecture **Block 2 - L1** in `accCalib.m`. Please add comments where necessary.

T.1.6 **(2 points)** Use the results of **T.1.5** and compose the transformation ${}^{\mathrm{e}}_{0}\mathbf{T}$ using $\mathbf{R}$ and $\mathbf{w}$ in the marked section of `main_acc_calib.m`. ${}^{\mathrm{e}}_{0}\mathbf{T}$ represents the coordinate system of the ellipsoid wrt. the base frame 0.

T.1.7 **(4 points)** Use the results of **T.1.5** and compose the homogeneous model matrix $\mathbf{T}_{\mathrm{m}}$ using $\mathbf{R}$ and $\mathbf{G}$ in the marked section of `main_acc_calib.m`. $\mathbf{T}_{\mathrm{m}}$ only contains the gains of the calibration and does not rotate the coordinate axes. Thus $\mathbf{T}_{\mathrm{m}}$ is approximately diagonal. Note that it is in general not sufficient to just paste $\mathbf{G}$ into $\mathbf{T}_{\mathrm{m}}$.

T.1.8 **(2 points)** Use the results of **T.1.5** and compose the homogeneous offset matrix $\mathbf{T}_{\mathrm{w}}$ using $\mathbf{w}$ in the marked section of `main_acc_calib.m`. $\mathbf{T}_{\mathrm{w}}$ only contains a translation.

T.1.9 **(2 points)** Use the results of **T.1.7** and **T.1.8** and compute ${}^{\mathrm{raw}}_{\mathrm{acc}}\mathbf{T}$ by concatenating $\mathbf{T}_{\mathrm{m}}$ and $\mathbf{T}_{\mathrm{w}}$ in the marked section of `main_acc_calib.m`. First apply the translation, then the scaling.

T.1.10 **(2 points)** Implement Equation 1 in the marked section of `main_acc_calib.m`. Check in the print out how much your calibration improves. This gives you some feedback if your calibration is correct.

T.1.11 **(2 point)** Finally, use ${}^{\mathrm{raw}}_{\mathrm{acc}}\mathbf{T}$ and ${}^{\mathrm{acc}}_{0}\mathbf{T}$ to calibrate and transform the raw measurements to the base frame 0 in the marked section of `main_acc_calib.m`. The script loads the measurements of the gravitational accelerations aligned to the different coordinate axes. You can verify if the print outs match your expectations.

Please submit `main_acc_calib.m` and `accCalib.m` containing your modifications for **T.1.3** to **T.1.11**.

1.3 Report (10 points)

R.1.1 **(2 points)** How many different poses do we need at least to get a good set of parameters for the implemented calibration algorithm?

R.1.2 **(2 points)** Why should linear accelerations be minimized while capturing data for the calibration? Explain and elaborate.

R.1.3 **(1 points)** How do you minimize linear accelerations?

R.1.4 **(1 points)** How do you minimize the influence of linear accelerations during calibration?

R.1.5 **(4 points)** Describe and explain the mathematical trick we use to get the model matrix $\mathbf{M}$ from a properly scaled ellipsoid matrix $\mathbf{A}_{\mathrm{fit}}$. Explain! Copying the formulas from the lecture script is NOT enough.